# Visualization of Single Perceptron

$$\underline{\mathbf{X}} = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ \dots \\ x_n \end{pmatrix} \Rightarrow \underline{\mathbf{X}} = \begin{pmatrix} 1 \\ x_1 \\ x_2 \\ \dots \\ \dots \\ x_n \end{pmatrix}$$

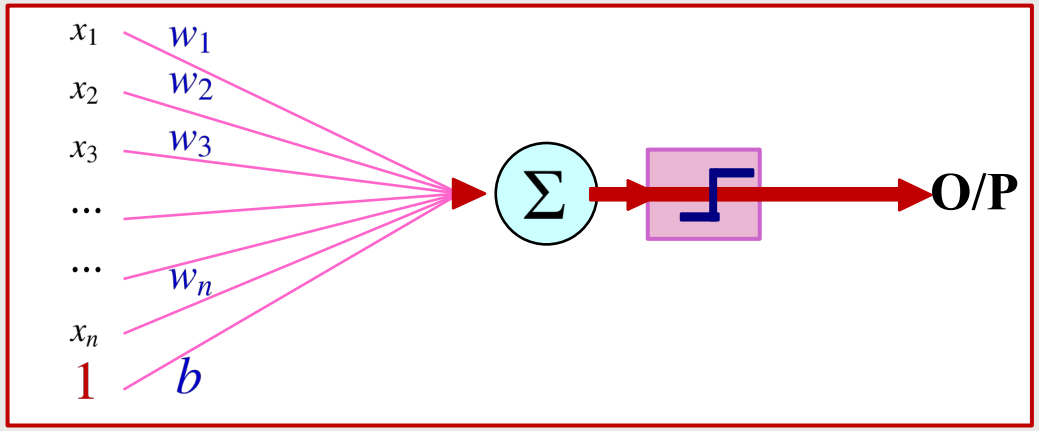
*Ugmented Vector*

$$\underline{\mathbf{W}} = \begin{pmatrix} b \\ w_1 \\ w_2 \\ \dots \\ \dots \\ w_n \end{pmatrix}$$

*Model Weights*

$$d(\mathbf{X}) = \underline{\mathbf{W}}^T \underline{\mathbf{X}} = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b * 1 \Rightarrow$$

$> 0 \Rightarrow$ Class I
$< 0 \Rightarrow$ Class II
$= 0 \Rightarrow$ Undefined Class




*Binary Classification (2 Classes)*